

# Jacen Li

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## Education

### University of Washington

Expected 2028

- BSECE - Firmware and Embedded Systems (GPA 3.88)

## Skills

**Languages:** C++/C++20, C, Assembly (x86-64, ARM), Python, Shell (Bash/Zsh), SVE, AVX2

**Tools:** Linux (Debian/Ubuntu), Codex/Claude Code, Git/Github, perf, GCC/Clang, GDB, objdump, Make/CMake

## Experience

### Arm

Summer 2026

*Software Intern (Incoming)*

Austin, TX

- Incoming software intern on platform performance tooling for Neoverse CSS HPC/AI platform
- Will work with Linux-based profiling, benchmarking, and PMU/perf-style measurement workflows

### National Institute of Standards and Technology

2023

*Sensor Science Division Intern*

Gaithersburg, MD

- Deployed a 61 node ADT7410 I2C temperature sensor network with Linux on Raspberry Pi to profile thermal gradients in the Tape Tunnel test chamber and automated calibration of legacy sensor networks.
- Quantified error propagation w/ Monte Carlo simulation, reduced temperature-induced uncertainty by 38%.

### Aeus Keyboards

2020 - 2025

*Founder & Operator*

[jtl06.github.io/keyboards.html](https://jtl06.github.io/keyboards.html)

- Founded custom keyboard company to >\$40K revenue; led design, manufacturing, and fulfillment
- Designed ATmega32U4 PCBs (KiCad) and wrote QMK firmware in C(matrix scan, debouncing, USB HID)

## Technical Projects

### BLT - Better Latency Tester (STM32)

[github.com/jtl06/BLT](https://github.com/jtl06/BLT)

- Built input-to-photon latency measurement device to measure and benchmark gaming PC responsiveness
- Utilized timers, circular DMA, and ISRs for us-level resolution, quantified DMA ISR jitter:  $\pm 8 \mu\text{s}$  at 2.5 kHz.
- Scripted OpenOCD + GDB flash and debug with Make; built Python GUI for data collection and CSV export

### SwiftFIX - SIMD FIX pre-parser for QuickFIX

[github.com/jtl06/SwiftFIX](https://github.com/jtl06/SwiftFIX)

- Hand rolled a C++20 FIX tag-value preparser that performs single-pass structural scanning, header indexing, body-length validation, and field-boundary extraction into a reusable zero-alloc offset table.
- Improved frame-splitting throughput from 146 MiB/s in QuickFIX to 1.60 GiB/s scalar and 2.19 GiB/s with AVX2, reducing cycles/message from 3,999 to 257 and nearly eliminating L1D misses.

### Libdivide - Fast Divide with ARM SVE

[github.com/ridiculousfish/libdivide/pull/158](https://github.com/ridiculousfish/libdivide/pull/158)

- Implemented ARM SVE support for libdivide's vector division paths for all widths with tests and docs.
- Benchmarked on Graviton4 showing ~5x faster 16-bit division vs NEON from wider scalable vectors/native lane ops, and ~3x faster 64-bit division vs NEON from SVE's native vector multiply-high instructions.

### Nycticorax

[github.com/jtl06/Nycticorax](https://github.com/jtl06/Nycticorax)

- Built a Discord agentic chatbot using Claude Code and Codex for iterative implementation and debugging
- Modularized codebase and maintained clear requirements/documentation so different AI tools and multi-agent routines could contribute coherently while managing limited context windows.
- Implemented selective memory, hybrid retrieval, and Postgres storage while controlling latency and cost

### PNVMe - FPGA NVMe device

[github.com/jtl06/PNVMe](https://github.com/jtl06/PNVMe)

- Built custom NVMe SSD controller on Artix-7 FPGA, enumerated as a PCIe NVMe Linux block device.
- Substantiated NVMe BAR and wrote custom XDMA to NVMe harness backed by onboard DDR3

### ARMv8 CPU

- Created 5-stage ARMv8 pipelined CPU with forwarding; benchmarked cache miss/branch CPI impact.